MILITARY SPECIFICATION

SYRINGE, IRRIGATING, DENTAL (Disposable)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sterile, disposable and plastic endodontic syringe suitable for flushing, irrigating and sterilizing root canals with medicated solutions.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th St., Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or be letterletter.

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SPECIFICATIONS

Federal

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Fiberboard.
PPP-B-676	- Boxes, Set-up.

Military

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-S-36586	- Sterilization Strip Test, Bacterial Spore.

STANDARDS

Federal

FED-STD-151	- Metals; Test Method.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIl-STD-794	- Parts and Equipment, Procedures for Packing of.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24513	- Syringe, Irrigating, Dental, Disposable, 6ml, Sterile

(Copies of specifications, standards, and drawing required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual, Stainless and Heat-Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI Z70.1 American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act (FDA) and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D.C. 20204).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. All material shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act. The portions of the unit that come in contact with the fluid to be injected shall be compatible with all known endodontic medications. Also, the plastic components of the syringe shall comply with USP requirements for Class III plastic containers.

3.1.1 Barrel. Syringe barrels shall be fabricated from rigid, transparent, virgin, nontoxic, high-grade polypropylene plastic (see 3.1). Selected regrind free from graduation filler and any other adulterants may be used. The barrel shall be considered to be transparent if the scale can be seen in daylight by the naked eye when viewed through the empty barrel from the side opposite to the scale.

3.1.1.1 Alcohol resistance of graduation markings. The graduation markings shall withstand vigorous rubbing of an alcohol moistened cotton swab and remain legible when tested according to 4.4.7.

3.1.2 Plunger. The plunger rod shall be fabricated from rigid polypropylene or polystyrene plastic. The plunger tip shall be fabricated from fresh stock, pharmaceutical grade, nonblooming, nonleaching, nontoxic, age-resistant, inert, natural rubber, or neoprene (see 3.1).

3.1.2.1 Inertness to medication. The plunger tip shall show no evidence of a reaction with a high pH buffer solution when tested according to 4.4.8.

3.1.3 Needle. The needle shall consist of a cannula, a plastic hub and a metal insert.

3.1.3.1 Cannula. The cannula shall be fabricated from either seamless drawn; or welded and drawn, stainless steel tubing, American Iron and Steel Institute (AISI) type 304 stainless steel. Material shall be tested, as specified in 4.4.5, for chemical composition.

3.1.3.2 Hub. The hub shall be made from a suitable plastic compound containing no elutable toxic materials. The hub shall be compatible with known medications.

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3.1.3.3 Metal insert. If a metal insert is used (to connect the hub to the cannula), it shall be fabricated from a suitable aluminum alloy (degreased and chemically passivated or anodized), or from stainless steel.

3.1.4 Lubricant. The lubricant on the exterior of the cannula shall be inert, non-toxic silicone compound suitable for the specific use. The silicone compound shall be pharmaceutical grade.

3.2 Design and construction. Shall be a sterile, pyrogen-free, non-toxic, disposable endodontic syringe. The syringe shall consist of a plunger, barrel and an endodontic needle, suitable for introducing medicated solutions to flush, irrigate and sterilize root canals. The syringe shall be packaged in an individual, sealed, sterile package as hereinafter specified. The sealed package shall maintain the sterility of its contents, and shall be so designed to permit ready dispensing of the syringe by sterile techniques.

3.2.1 Barrel. The syringe barrel shall be fabricated from plastic, as specified in 3.1.1. The barrel shall consist of a tube closed at one end, except for an orifice, and open at the other end to receive the plunger. At the closed end, there shall be formed a tapered tip, which shall have a 0.060 inch taper per inch, to receive a standard endodontic needle. The tip shall be formed concentric to the axis of the barrel, and shall make a mechanically stable, fluid tight connection (see 3.4) with an endodontic needle having a luer hub with hub cavity in accordance with Figure 2 (female end) of American National Standards Institute (ANSI) Standard Z70.1.

The tip at the distal end shall have an outside diameter of 0.155 to 0.160 inch. The outside diameter of the tip at an altitude of 0.250 inch from the distal end shall be 0.170 to 0.175 inch. The length of the tip, except for a possible undercut or cylindrical cut at the junction with the barrel, may be greater than 0.250 inch providing the same 0.060 inch/inch taper is maintained. The wall at the open end of the barrel shall extend radially outward to form a flange of suitable size and shape to provide finger grip and anti-roll features. The inside of the barrel shall have a very fine, uniform surface, substantially free from pits, grind marks, bubbles, mold marks, flash, open air lines, or other defects that would affect the fluid tightness and smoothness of its fit with the plunger. The inside of the barrel shall be lightly coated with the silicone lubricant specified in 3.1.4. Dimensions of the barrel shall be as specified on DPSC Drawing 24513.

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Inside diameter shall be such as to provide accurate fit with plunger tip, resulting in a fluid-tight fit (see 3.2.2 and 3.4.2), and yet permitting plunger to move freely throughout its excursion in the barrel. Ungraduated portion of barrel, at proximal end, shall be of sufficient length to permit plunger tip to be aligned with top graduation mark, while barrel is encompassing entire rubber tip, but shall be not greater than one-third of the overall length of the syringe barrel.

3.2.1.1 Graduations and other markings. Barrel shall be calibrated to 6 ml (cc) with main graduation at each 1 ml (cc), subgraduations at each 0.2 ml (cc) and numbered at each 1 ml (cc). The outside surfaces of the barrel shall be marked with a graduation scale and the name or registered trademark of the contractor. The graduation scale shall be applied to the barrel in such a manner as to be approximately in line with the flattened side of the flange. The zero (0) mark of the scale shall register precisely with the distal ring of the plunger tip, when plunger is pushed to the bottom of the barrel. The scale and numbering shall be in cubic centimeters. Main graduation marks shall be at least 33-1/3 percent longer than subgraduation marks. No number is required at the zero (0) mark. The markings shall be legible and of a dark, contrasting color, easily readable.

3.2.2 Plunger. The plunger shall be either hollow or solid, fabricated from the material specified in 3.1.3. The distal end of the plunger shall be shaped to permit attachment of a plunger tip. The proximal end of the plunger shall extend well out of the barrel, and shall be shaped to form a knob of thickness and diameter appropriate to the size of the syringe, and shaped to allow a suitable finger grip for usage. The distal end of each plunger with its attached plunger tip shall accurately fit its individual barrel so as to make the syringe successfully withstand the fluid tightness requirements specified in 3.4.2, permit the plunger and plunger tip to draw liquids, and yet move freely and smoothly throughout its excursion in the barrel. The plunger tip shall be securely attached to the plunger rod so as not to become loose or detached when in use. The tip shall be shaped to closely fit the distal end of the barrel so as to remove all fluid from the inside of the barrel when the plunger is pushed all the way in. In addition, the tip shall be designed, e.g., beveled, in order to permit easy and efficient removal of air bubbles from fluid path. The plunger tip shall be properly treated with the silicone lubricant specified in 3.1.4. An excess of lubricant that detracts from appearance, interferes with visibility of plunger tip ring, or can be caused to form pools or droplets, or can be washed out with the solution being used, shall be cause for rejection.

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Graduation markings, denoting correct, accurate capacity, shall register precisely with the first ring at the plunger tip distal end, touching sides of the barrel. Tip shall be of a permanent, suitable color so that the first ring at the plunger tip distal end, touching the sides of the barrel, shall be clearly visible and shall accurately indicate through the barrel wall. Each plunger shall have a suitable solid thumbrest at its proximal end. Thumbrest shall be perpendicular to the centerline of the plunger and may be flat or slightly curved. As an alternate, thumbrest may have a centered hole which shall not exceed 3/8 inch in diameter. Thumbrest shall be fabricated from material as specified in 3.1.2.

3.2.2.1 Plunger tip resiliency. The rubber plunger tip when tested as specified in 4.4.9.3 shall not crack, crumble or become fissured. Furthermore, after a one-minute recovery time, it shall again be circular in appearance. An obviously oval shape indicating a lack of adequate recovery is cause for rejection.

3.2.3 Needle.

3.2.3.1 Cannula. The cannula shall be a straight tube with a special notch at the distal end. The point shall be ground flat. Both outer and inner edges of the point shall be free from burrs and slivers. The proximal end of the cannula shall extend well into the metal insert sufficiently to allow for secure staking. The cannula and metal insert shall be centered in the distal end of the hub in such a manner as to provide a straight continuous fluid path from the hub cavity. The cannula shall not deviate from the central longitudinal axis of the needle by more than 3 degrees. The exterior of the cannula shall be clean, smooth and free from pits, deep furrows and die marks. Dimensions of the cannula and the special notch shall be as specified on DPSC Drawing 24513.

3.2.3.1.1 Corrosion resistance of cannula. When tested as specified in 4.4.6, cannula shall show no appreciable corrosion.

3.2.3.2 Hub. Shall be Luer-lock hub to provide means for facilitating attachment to, and removal from, a syringe so designed and constructed to form a leak-proof connection with male Luer lock connection constructed in accordance with Figure 4 of American National Standards Institute (ANSI) Z70.1, and shall make a mechanically stable, fluid tight connection (see 3.4) with a syringe having a Luer lock tip. The tapered portion of the hub cavity shall be 0.060 inch per inch in accordance with Figure 1 of ANSI Z70.1.

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The hubs shall be tested for diameter and depth of hub cavity as specified in 4.4.1. Hubs which do not fit the plug gage within the "GO" -"NO GO" limits specified in 4.4.1 shall be considered not acceptable. The fit of the Luer taper of plastic hubs on the plug gage need not be tested with the feeler gage present. When the hub is fitted onto the plug gage as specified in 4.4.1, there shall be no movement of the hub on the gage when the hub is pressed lightly from side to side in all directions. Hub and cannula shall be securely assembled. Soft solder (lead-tin alloy) shall not be used in joining the cannula and hub.

3.3. Accuracy. When tested as specified in 4.4.3, syringes shall deliver plus or minus 5 percent of the capacity indicated at each graduation mark tested.

3.4 Fluid tightness. The syringe shall be tested for fluid tightness between the plunger and barrel, and between barrel tip and needle hub as specified in 4.4.9.1.

3.4.1 Suction. There shall be no air leakage as revealed by the suction test as specified in 4.4.9.2.

3.4.2. Plunger fit. The plunger shall make a snug fit in the barrel to provide leak-tight operation. At no part of the travel of the plunger shall the plunger fit be so loose as to permit movement of the plunger by its own weight when tested as specified in 4.4.9.4.

3.5 Freedom from foreign matter. When tested as specified in 4.4.2 the syringe (prior to sterilization) shall be free from chips and other foreign matter. In addition, a visual examination of the lumen of the cannula shall reveal no presence of foreign matter, chips, burrs, or occlusion of the lumen.

3.6 Stiffness. When tested as specified in 4.4.12, the maximum deflection of the cannula shall not exceed 0.033 inch.

3.7 Security of staking. When tested as specified in 4.4.11, the cannula shall not loosen or pull out of the hub.

3.8 Resistance to breakage. When tested as specified in 4.4.10, there shall be no breakage.

3.9 Freedom from pyrogens. When tested as specified in 4.4.4, syringes shall comply with the pyrogen test requirements of the USP.

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3.10 Sterility. The syringes shall be sterile when tested with a biological indicator as specified in 4.4.13.

3.10.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item Identification.
b. National Stock Number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-RSQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA, 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.10.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.11 Safety (toxicity) and lead content.

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3.11.1 Safety. When tested as specified in 4.4.14.1 and 4.4.14.4 there shall be no deaths or discernible symptoms of toxicity of the test animals in determining acute toxicity or adverse tissue reaction. The injection sites on the test animal shall be free from evidence of irritation or toxicity (trauma shall be ruled out with suitable saline control sites).

3.11.2 Lead content. When tested as specified in 4.4.14.2 or 4.4.14.3, all plastic and rubber components coming in contact with the medication being administered shall contain not more than 5 parts per million (p.p.m.) of lead.

3.12 Certificates. For destination, inspection only, certificates of freedom from pyrogens, and safety, as specified below, shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-RSQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificates of freedom from pyrogens and safety shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.12.1 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens, as specified in 3.8 and 4.4.7, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.12.2 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.11 and 4.4.14, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.13 Instructions. Each unit package shall be provided with instructions describing the proper techniques for dispensing the sterile syringes, as applicable, from the containers and preparing them for use. In addition, information as to methods for destroying syringes, as applicable, to prevent their re-use shall also be shown.

3.14 Freedom from loose or detachable flash. When tested as specified in 4.4.2.1, flash at the fluid path of the syringe shall not be detachable.

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3.15 Workmanship. Syringes shall be free from defects which detract from their appearance or impair their serviceability.

3.16 Precautionary information (plastic hub needles only). For needles with plastic hubs, each unit package shall be provided with information cautioning the user against the use of parenterals, solution, etc. that would be harmful to the plastic.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Governments request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic and metals may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105 and, unless otherwise specified, shall be based on the unit of product being one syringe (end item).

4.2.2. Sampling for examination. A sample of syringe shall be selected from each lot in accordance with MIL-STD-105. The inspection level and acceptable quality level (AQL) shall be as indicated in table I.

TABLE I. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major	I	1.0
Minor	I	4.0

4.2.3 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and as indicated in table II.

TABLE II. Sampling criteria for tests

Component	Characteristics	Requirements	Test procedure	Inspection-level	AQL(percent defective)
Barrel	Graduation markings	3.1.1.1	4.4.7	S-1	2.5
Plunger tip	Inertness	3.1.2.1	4.4.8	S-2	0.25
Plunger tip	Resiliency	3.2.2.1	4.4.9.3	S-3	0.25
Barrel	Dimensions	3.2.1	-	S-1	1.0
Needle		3.2.3	-	S-1	1.5
Cannula	Composition	3.1.3	4.4.5	S-1	1.5
Cannula	Corrosion resistance	3.2.3.1.1	4.4.6	S-2	1.0
Barrel Plunger	Lead content	3.1.1.2	4.4.14.2 or 4.4.14.3	S-2	1.0

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TABLE II. Sampling criteria for tests (continued)

Component	Characteristics	Requirements	Test procedure	Inspection level	AQL (percent defective
Needle	Gaging	3.2.3.2	4.4.1	S-1	1.0
Needle	Resistance breakage	3.8	4.4.10	S-1	1.5
Needle	Security of staking	3.7	4.4.11	S-1	1.5
Needle	Stiffness	3.7	4.4.12	S-1	1.5
Needle	Safety	3.11.1	4.4.14.4	S-1	1/
Syringe W/O needle	Flash	3.14	4.4.2.1	S-2 2/	1.0
Syringe	Freedom of foreign matter	3.5	4.4.2	S-3	1.5
Syringe	Freedom from pyrogens	3.9	4.4.4	S-1	1/
Syringe	Accuracy	3.3	4.4.3	S-1	2.5
Syringe	Suction test	3.4.1	4.4.9.2	S-2	1.0
Syringe	Pressure test	3.4	4.4.9.1	S-2	1.0
Syringe	Sterility	3.10	4.4.13	S-3	1/
Syringe	Safety	3.11.1	4.4.14.1	S-1	1/
Syringe	Plunger fit	3.4.2	4.4.9.4	S-3	0.25

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1/ Acceptance number shall be zero for all sample sizes.

2/ Sample shall be selected at random from those syringes with flash found in the sample group used in the examination for Classification of Defect 124.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more characteristics.

4.3 Examination. The syringes shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any syringes in the sample containing one or more defects shall be rejected and, if the number of defective syringes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Needle missing or dislodged.
102	Cannula lumen not free of obstructions or foreign matter.
103	Cannula lumen not free of burrs and slivers.
104	Cannula outer surface not smoothly finished or free of pits, tool marks, burrs, scale, or corrosion.
105	Cannula not free of cracks.
106	Cannula not straight.
107	Cannula exterior not free from foreign matter.
108	Hub not free of cracks or flash.
109	Hub not free of burrs, slivers, or foreign matter.
110	Point blunt or bent, 1/ not sharp, not symmetrical or properly formed, or not free of burrs, feathered edges, or other detrimental defects (examined under 7X magnification).
111	Plunger rod, plunger tip, or both, missing.
112	Plunger or plunger tip frozen in barrel.
113	Inside of barrel not coated with required lubricant.
114	Inside of barrel contains excessive lubricant.
115	Barrel tip orifice missing or blocked.
116	Graduation markings incomplete, missing, illegible, or incorrect.

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TABLE III. Classification of defects (Continued)

Categories	Defects
Major	
117	Barrel not clear, plunger tip not clearly visible making accurate alignment with graduation markings difficult or impossible.
118	Taper of barrel tip not smoothly finished, free of pits, tool marks, bubbles, or loose or detachable flash and burrs.
119	Inside surface of barrel not fine uniform surface, free from pits, grind marks, bubbles, mold marks, flash, open air lines, or other defects that would affect fluid tightness and smoothness of its fit with the plunger.
120	Plunger tip not smooth, free from flash, bubbles, breaks, cracks, or other defects affecting its fit with plunger.
121	Plunger does not operate smoothly in barrel.
122	Plunger tip detaches from plunger rod when pushed back and forth in syringe barrel.
123	Barrel or plunger (including tip) not free of cracks, breaks, crazing or tackiness.
124	Bore of syringe tip not free from flash which obstructs bore of tip below the minimum area.
Minor	
201	Deep scratches (not interfering with reading of dosage, see 117) on outside surface of barrel (except on tapered portion of tip, see 118).
202	Discoloration of outside surface of hub.
203	Contractor's name or registered trademark, missing, incomplete, or not legible.
204	Graduation scale not approximately in line with flattened side of flange.
205	Instructions missing, incomplete, or not legible.

1/ A bent point is defined as any deviation of the point from the longitudinal axis of the needle when viewed at 7X from the side of the needle and in a line parallel to the plane of the bevel.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Diameter and depth of hub cavity. Hub cavity shall be tested on the male Luer taper gage in accordance with plug gage of Figure 5 of ANSI Z70.1. (This gage shall be as shown in ANSI Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.9). The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with Figure 6 and 8 of ANSI Z70.1. The fit of the Luer taper of plastic hubs on the plug gage need not be tested with the feeler gage present. All hubs shall provide a firm, non-wobble fit when testing the Luer taper of hubs on the plug gage. The flange on the Luer hub shall be properly recessed from the proximal edge and of dimensions to provide a full, firm, leak-proof fit of the hub and tip (see figure 3 of the ANSI Standard Z70.1).

4.4.2 Freedom from foreign matter test. The hub cavity shall be examined under 7X magnification, and good illumination, for chips and other foreign matter. In addition, a cotton swab tightly wrapped on the end of a small wooden applicator shall be turned down into the bottom of cavity, withdrawn, and examined for chips and other foreign matter under 7X magnification and good illumination. The cannula shall be tested for blockage by inserting a wire through the lumen of the cannula. The syringe shall be filled with distilled water by pouring the water into the barrel. Insert the piston and eject the distilled water into a medium texture, quantitative filter paper properly inserted into a 60°, 75 mm, smooth glass funnel. As a blank, take an equal amount of distilled water and pour it into a similar, separate funnel filter paper arrangement as used for the syringe. Examine the residue on the filter papers at 7-power magnification, after the water has passed through them, and determine if foreign material ascribable to the needle is present.

4.4.2.1 Freedom from loose or detachable flash. Flash at the bore of the tip shall be flexed with a blunt probe, of no more than .015 inch diameter, to determine that flash, if present when examined at 7X, will not become detached in use. Flexing shall be for three cycles. Each cycle is one back and forth full flexure.

4.4.3 Accuracy test. Syringes shall be tested for accuracy of delivery at each main graduation mark (see Table II) using water and a calibrated burette, or as an alternate, gravimetrically taking into account the temperature and density of water. The first ring from the distal end of the plunger tip shall be aligned with each graduation mark tested. One-cc syringes shall not be tested for accuracy at the first 15% of the scale.

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4.4.4 Freedom from pyrogens test. Place 10 syringes in a sterile, pyrogen-free flask, pour 250 milliliters (ml.) of sterile, pyrogen-free saline test solution (USP) into the flask and then heat the contents of the flask at not less than 40° Centigrade (C.) nor higher than 50° C. for 1 hour. The resulting saline extract shall be submitted to the "Pyrogen Test" as described in the USP.

4.4.5 Chemical analysis. Cannula material (see 3.1.3.1) shall be subject to chemical analysis in accordance with method 111 or 112 of FED-STD-151.

4.4.6 Corrosion-resistance of cannula. The cannula shall be immersed in 10 percent citric acid at room temperature for 5 hours and, after removal from the acid, shall be boiled in distilled water for 30 minutes, and then shall remain immersed for 48 hours in the distilled water. At the end of the 48-hour period, the cannula shall be allowed to dry by evaporation, and shall then be examined for evidence of corrosion. A glass vessel shall be used throughout this test.

4.4.7 Permanency of graduation markings. The graduation markings shall withstand five vigorous rubbing cycles of a 70% isopropyl alcohol moistened cotton swab. A cycle shall be one up and one down stroke along the length of the barrel. The markings shall remain legible.

4.4.8 Plunger tip inertness. The rubber plunger tip shall be removed from the plunger rod, wiped with a clean, dry cloth to remove silicone lubricant, and immersed in a test tube under a volume of pH 10.5 buffer solution equal to the capacity of the syringe. The exposure shall be for one hour at 68° to 77°F, (20° to 25° C). There shall be no color or any other visible change of the buffer solution. A control shall be used during the exposure consisting of the same amount of buffer solution in a test tube which does not contain a plunger tip.

4.4.9 Fluid tightness test. The fluid tightness shall consist of the pressure test and the suction test described in the following paragraphs:

4.4.9.1 Pressure test. The water pressure test system shall have a water pressure gage and a medication injection plug. The needle of the syringe shall be inserted into the medication injection plug and the water pressure brought up to 45 psig. The plunger shall then be withdrawn to the 50% point of the graduated capacity where it shall be allowed to remain for 10 seconds. The plunger shall then be withdrawn to the 75% point and held there for another 10 seconds.

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The plunger shall then be withdrawn to the 100% point for a final 10 seconds. During each 10-second period there shall not be enough water to pass the connection between the needle hub and Luer tip of the barrel or needle-barrel junction and between the plunger tip and barrel of the syringe to produce a falling drop. After each successive test, any water that may have passed shall be wiped with a dry cloth and the next test applied. The needle of the syringe shall be withdrawn from the medication injection plug and the water ejected at the rate of 1/2 cc/second until the plunger reaches the zero mark. The needle shall not leak or separate at its junction with the barrel.

4.4.9.2 Suction test. An amount of water equal to more than one-half the capacity shall be drawn into the syringe through the endodontic needle. The syringe shall then be held vertically, needle up, to allow air to rise above the water and be expelled. Tapping the barrel of the syringe will facilitate removal of trapped air. The plunger shall be pressed inward to the 50% mark. A soft rubber stopper shall then be used to seal the needle. The plunger shall then be carefully withdrawn, at a rate of about one inch/10 seconds, to the full mark. After the plunger has reached the full mark, there shall be no more than three bubbles coming by the distal ring for a least 10 seconds. Should there be more than three air bubbles passing the distal ring during the 10-second period, there shall be an additional period of observation of 15 seconds during which there shall be no bubbles passing the distal ring. Upon release of the plunger, it shall return to the 50% mark.

Air bubbles which appear on the distal surface of the plunger tip during the withdrawal of the plunger shall be disregarded. Some air bubbles frequently appear under conditions of reduced pressure because of phenomena such as decreased solubility of air in water, interfacial tension, and aggregation of minute bubbles into larger ones.

4.4.9.3 Plunger tip resiliency. The unmounted rubber plunger tip shall be free of visible lubricant. If necessary wipe dry with a cloth. The outside diameter shall be measured to the nearest 1/16 inch. The plunger tip shall then be compressed laterally to 7/16 to 1/2 of the outside diameter for 15 seconds. This shall be done in a vise the jaws of which have smooth faces. The compressed plunger tip shall be observed for cracks, crumbling or fissures. The plunger tip is removed from the vise and allowed to recover for 60 seconds after which it is examined for circularity.

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4.4.9.4 Plunger fit. The syringe shall be held vertically with the Luer tip exposed. The plunger shall be worked back and forth twice, for the full length of the barrel. The plunger shall be withdrawn from zero, in increments of 1/10 the capacity, and observed for 10 seconds at each level. The Luer tip can be up or down in this test. Movement of the plunger by its own weight is cause for rejection. The same test shall be used for syringes with permanently attached needles except that 20 seconds observation time shall be used.

4.4.10 Resistance to breakage test. The needle hub shall be held in a fixed position, and a force shall be applied to the cannula, at a distance from the hub of 5/8 inch, in order to bend the cannula in a clockwise direction through an arc of 25 degrees from the original or neutral position. The operation shall then be reversed (i.e., bent through an arc of 25 degrees from the neutral position in a counter-clockwise direction), and finally, the cannula shall be returned to neutral position. This complete operation (or cycle) shall be repeated 20 times.

4.4.11 Security of staking. A straight pull of 5-pounds shall be applied between the cannula and hub for a period of 1-minute.

4.4.12 Stiffness test. The cannula shall be placed across two test edges (0.002 inch radius maximum) separated by a distance of 0.625 plus or minus 0.003 inch. A load of 40 plus or minus 1/4 ounces shall be applied at the center of the span and the resulting maximum deflection of the cannula shall be measured using a gage with graduations of at least 0.001 inch. The force shall be applied through an edge with a maximum radius of 0.036 inch, and centered to within ± 0.003 inch.

4.4.13 Sterility Test.

The syringe shall be tested for sterility in accordance with the method specified in the U.S.P. using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium for the paper spore strip (carrier) shall be soybean-casein digest. The incubation media for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30° to 35° C for one half of the specimens and soybean-casein digest at 20° to 25° C for the remaining half of the specimens. The minimum incubation time shall be ten days. **Sampling shall be as follows:**

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Medium	Paper Spore Strip	Inoc. Prod. or Inoc. Simulated Prod.
Soybean-casein digest	S-3*	S-3*
Fluid thioglycollate or Alterative thioglycollate		S-3*

*Use a minimum of 20 specimens. Acceptance number is zero.

4.4.13.1 Inoculated product or simulated product. For each type of sterzilization, the organism used shall be as listed below:

Type of Sterilization	Organism
Steam	B. stearothermophilus
Dry heat	B. stearothermophilus B. subtilis
Ethylene oxide	B. subtilis
Gamma radiation	B. pumilus

The number of viable spores shall be sufficient to provide assurance that the product has been sterilized.

4.4.13.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586A, Sterilization Test Strip Set, Bacterial Spore. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.13 Sterility test and 4.4.13.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.13.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P. and paragraphs 4.4.13 and 4.4.13.1.

4.4.13.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

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4.4.13.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the U.S.P. The inspection level shall be as stated in paragraph 4.4.13 above.

4.4.14 Safety (toxicity) and lead content tests.

4.4.14.1 Safety (toxicity). The finished syringes shall be tested for safety in accordance with the current edition of the USP, for Transfusion and Infusion Assemblies, except that a minimum of 5 samples shall be used to prepare the extract. Where more samples are indicated by Table II they shall be used in multiples (or near multiples) of 5 to prepare a single extract.

4.4.14.2 Test for lead. Ten (10) gms of rubber or plastic shall be taken, ignited to char, and then further ignited in a muffle furnace of 700°C. to a white ash. The ash shall be boiled for 15 milliliters (ml) of 10 percent nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the Pharmacopeia of the United States, shall be applied using a control of 5 p.p.m. or 12 p.p.m. of lead solution as given in the USP test. If the sample solution is darker than the limit, small aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in p.p.m. is determined.

4.4.14.3 Alternate test for lead. Accurately weigh a 5-gram sample into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnance at 650°C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deinoised water and dilute to 50 ml. Prepare standard lead solutions of 1, 2 and 5 p.p.m. lead.

4.4.14.4 Needles with plastic hubs. Needles furnished with plastic hubs shall be tested for safety as follows:

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Cover 20 needles with 30 ml. of sterile, pyrogen-free saline test solution (USP) and warm at 37°C. for 24 hours. Decant the saline and test, undiluted, for both acute toxicity and tissue sensitivity. Determine acute toxicity by injecting 1.0 ml. of the eluate intravenously into each of 10 nonfasted, healthy 28 to 32 - gram mice. Observe the animals at 4, 24, and 72 hours after injection for any deaths or discernible symptoms of toxicity. Determine tissue reaction by injecting 0.20 ml. of the eluate into each of 10 sites on the back of a rabbit; the injections shall be made in such a manner as to raise a distinct bleb. Observe the injection sites at 4, 24, and 48 hours after injection for any evidence of redness, swelling, or hardness. Trauma shall be ruled out with suitable saline control sites.

4.5 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Individual package. Each syringe shall be packaged in a sealed peel-open type container or in a rigid plastic tamperproof container capable of protecting the needle cannula and maintaining sterility of the contents unless the package is damaged or opened. The container shall be so designed to facilitate dispensing of the syringe by sterile techniques.

5.1.1.2 Unit package. Fifty individually packaged syringes shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. Syringes shall be individually packaged as specified in 5.1.1.1 and further packaged level C in accordance with the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

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5.2.1 Level A. Five unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Five unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for syringes shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Level A and B. Syringes packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Syringes packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B and C.

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determined valid for use in acquisition have been reviewed an

5.3.1.1 Individual container. Labeling shall meet the requirements of the Federal Food, Drug and Cosmetic Act. In addition to any commercial markings, each individual container shall also be marked with the item identification, name and address of the manufacturer or registered trademark, sterility lot (control) number, syringe and needle size, instructions for use and destruction, and the legend: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement. As an alternate, the instructions for use and destruction may be shown on the unit package. When the registered trademark of the syringe is visible through the carrier of the individual container, it shall not be required on the individual container.

5.3.1.2 Unit package. Each unit package shall be marked as specified in MIL-STD-129. Marking shall include the lot number. The instructions for use and destruction as specified in 3.13 need not be shown on the unit package when shown on the individual container.

5.3.1.3 Exterior container. Each exterior container shall be marked as specified in MIL-STD-129. Marking shall include the lot number.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number coverage. The following National Stock Number is covered by this document:

Stock Number	Item Identification
6520-01-036-0723	SYRINGE, IRRIGATING, DENTAL, Disposable, 6 ml, 50s: Sterile.

6.3 Supersession data. This specification includes the requirements of Medical Procurement Item Description No. 1, dated 31 May 1985, for NSN: 6520-01-036-0723, SYRINGE, IRRIGATING, DENTAL, Disposable, 6 ml, 50s: Sterile.

6.4 Subject term (key word) listing:
Dental Syringe
Disposable Syringe
Irrigating, 6 ml
Sterile

MILITARY INTERESTS:

Custodians

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:

VA-OSS
PHS
FDA-MPQAS

48978

PREPARING ACTIVITY:

DoD - MB

AGENT:

DLA-DM

Project No. 6520-1868

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-42050	SYRINGE, IRRIGATING, DENTAL (Disposable)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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